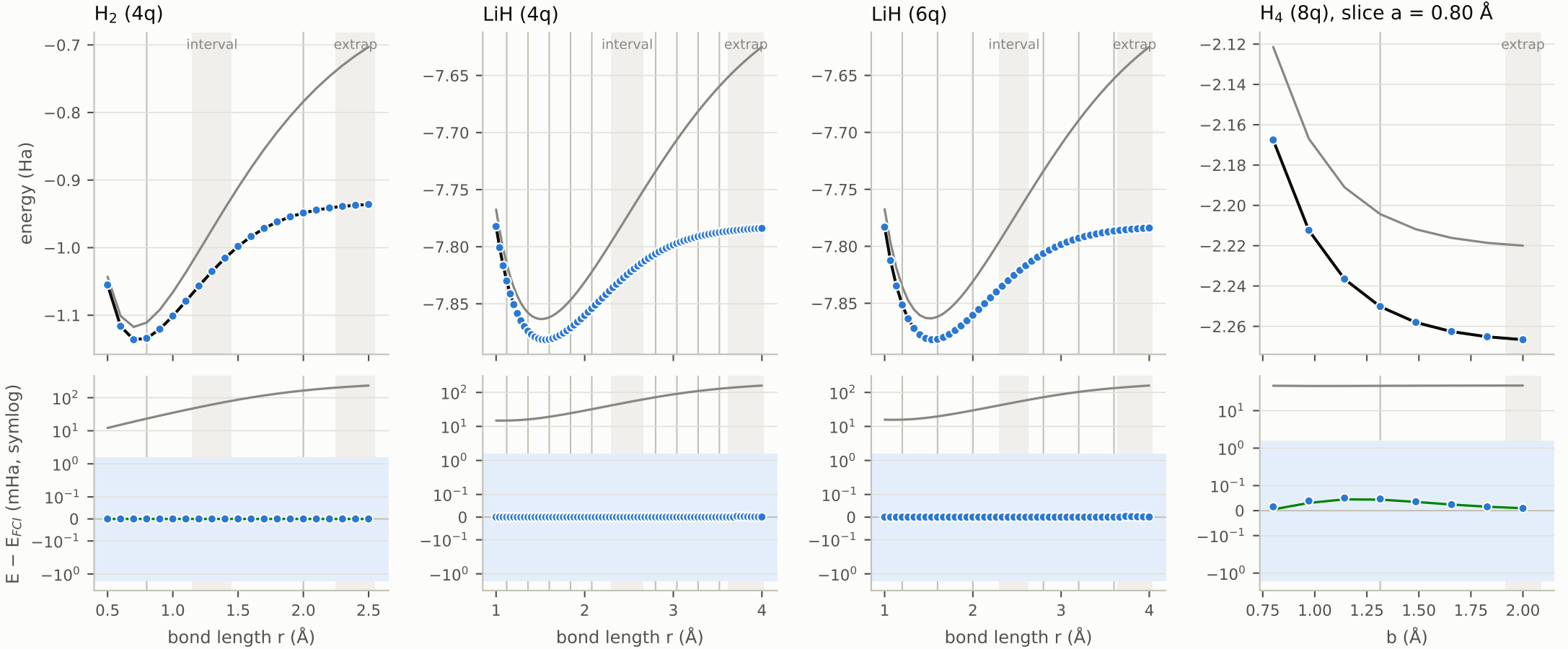Potential-energy curves and residual vs FCI